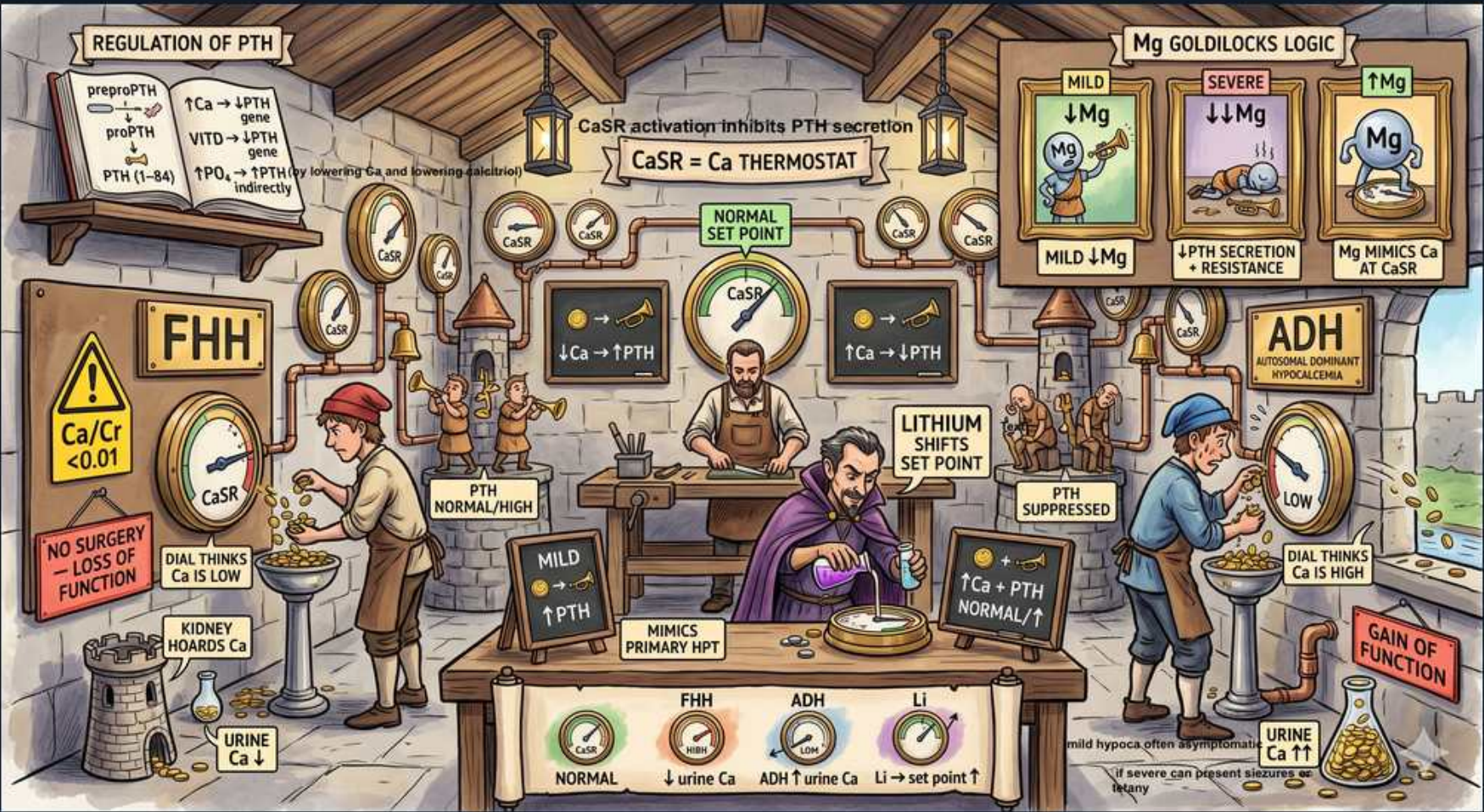


Calcium/PTH/VitD — The CaSR Thermostat Workshop





## 1. PTH regulation recap

### WHAT YOU SEE

Top-left panel showing prepro-PTH to proPTH to PTH(1-84) processing and regulators.

### WHAT IT ENCODES

PTH is synthesized as prepro-PTH, cleaved to proPTH, then to the active 84-amino-acid hormone PTH(1-84), which has a short half-life (~2-4 min) for tight minute-to-minute control. Secretion is RAISED by low ionized calcium and low calcitriol, and is also stimulated by high phosphate (partly indirectly, by binding calcium and lowering ionized Ca). Calcitriol feeds back to suppress PTH gene transcription — the rationale for vitamin D analogs in secondary hyperparathyroidism of CKD.

### BOARD TRAP

A stem says calcitriol STIMULATES PTH secretion — WRONG. Calcitriol provides NEGATIVE feedback, suppressing PTH synthesis. The discriminator: low Ca and low calcitriol drive PTH up; high Ca and high calcitriol turn PTH down.

## 2. CaSR = Ca thermostat

### WHAT YOU SEE

Central banner: CaSR is a thermostat with a normal set point; activation inhibits PTH secretion.

### WHAT IT ENCODES

The calcium-sensing receptor behaves like a thermostat: its SET POINT is the ionized calcium concentration at which PTH secretion is half-maximal. ACTIVATING the CaSR (rising ionized Ca, or agonists like cinacalcet, or hypermagnesemia) SUPPRESSES PTH. The calcimimetic cinacalcet exploits this to lower PTH in secondary/tertiary hyperparathyroidism and in parathyroid carcinoma. Mnemonic: 'sensor sees high Ca → shuts PTH.'

### BOARD TRAP

A stem implies that high calcium ACTIVATES the parathyroid to release more PTH — WRONG. CaSR activation by high calcium INHIBITS PTH secretion (negative feedback). The discriminator: the CaSR is an off-switch for PTH when calcium is high; it is not a stimulatory sensor.

## 3. FHH warning sign

### WHAT YOU SEE

Yellow hazard sign labeled FHH with Ca/Cr clearance ratio <0.01.

### WHAT IT ENCODES

Familial hypocalciuric hypercalcemia (FHH) is an autosomal dominant INACTIVATING CaSR mutation. The receptor under-senses calcium, so the setpoint shifts up: the body tolerates mild lifelong hypercalcemia with normal-to-mildly-high PTH. The hallmark distinguishing it from primary hyperparathyroidism is INAPPROPRIATELY LOW urinary calcium, with a calcium/creatinine clearance ratio <0.01. It is usually asymptomatic and benign.

### BOARD TRAP

A young patient with mild hypercalcemia, family history, and normal/high PTH is diagnosed with primary hyperparathyroidism and sent to surgery — WRONG if urine Ca/Cr clearance is <0.01. That ratio indicates FHH. The discriminator: primary HPT typically has urine Ca/Cr ratio >0.02 (normal/high urine calcium); FHH has <0.01 (low urine calcium). Always check urine calcium before parathyroidectomy.

## 4. No surgery / loss of function

### WHAT YOU SEE

A red 'NO SURGERY' stamp next to 'loss of function'.

### WHAT IT ENCODES

FHH is benign and managed conservatively with reassurance and monitoring — no calcium restriction, no surgery. Parathyroidectomy fails because the defect is a generalized CaSR setpoint problem in every parathyroid cell (and the kidney), not a discrete adenoma; removing glands leaves residual tissue that resets to the same abnormal setpoint and risks hypoparathyroidism.

### BOARD TRAP

A patient with confirmed FHH (low urine Ca/Cr) undergoes parathyroidectomy for 'hypercalcemia' — WRONG and harmful. Surgery does not cure FHH because the problem is the CaSR setpoint, not a parathyroid tumor. The discriminator: FHH = genetic, lifelong, asymptomatic, low urine Ca → observe; primary HPT with end-organ effects → consider surgery.

## 5. Kidney hoards Ca / urine Ca low

### WHAT YOU SEE

A tower hoarding calcium with a urine arrow pointing down.

### WHAT IT ENCODES

In FHH the inactivated CaSR in the renal thick ascending limb fails to sense calcium, so the kidney keeps reabsorbing calcium even when serum calcium is high — producing the characteristic INAPPROPRIATELY LOW urinary calcium. (Normally, a high serum calcium activates the renal CaSR to promote calciuria; that brake is broken in FHH.) This renal handle, independent of PTH, is what creates the low Ca/Cr clearance ratio.

### BOARD TRAP

A stem attributes the low urine calcium of FHH purely to high PTH driving renal reabsorption — WRONG/incomplete. The low urine calcium stems from the defective RENAL CaSR itself (PTH-independent reabsorption), which is why FHH urine calcium stays low even though PTH is only normal/mildly elevated.

## 6. CaSR dial set point

### WHAT YOU SEE

Central craftsman adjusting a dial; arrows showing low Ca raises PTH and high Ca lowers PTH.

### WHAT IT ENCODES

The setpoint is the calcium concentration at which PTH secretion is half-maximal; the steep sigmoid relationship means small calcium changes near the setpoint produce large PTH swings. Shifting the dial shifts the regulated calcium: inactivating CaSR (FHH) raises the setpoint (tolerates higher Ca), activating CaSR (ADH) lowers it (defends a lower Ca), and lithium nudges it upward pharmacologically.

### BOARD TRAP

A stem treats the setpoint as fixed and unaffected by drugs — WRONG. Lithium raises the setpoint and calcimimetics (cinacalcet) lower the effective setpoint, both shifting PTH output without any parathyroid pathology. The discriminator: medications can move the CaSR dial, so always review the drug list before invoking primary gland disease.



## 7. Lithium shifts set point

### WHAT YOU SEE

A purple wizard labeled Lithium shifting the thermostat set point upward.

### WHAT IT ENCODES

Lithium raises the CaSR setpoint (the parathyroid needs a higher calcium to shut off), so chronic lithium causes mild hypercalcemia with normal-to-high PTH and reduced urinary calcium — a near-perfect mimic of primary hyperparathyroidism (and of FHH). It can also cause true parathyroid hyperplasia over time. Management is to review/adjust lithium and monitor; surgery is generally avoided unless clearly indicated.

### BOARD TRAP

A bipolar patient on lithium with mild hypercalcemia and high-normal PTH is rushed to parathyroid surgery — WRONG before reviewing the medication. Lithium-induced hypercalcemia mimics primary HPT. The discriminator: take a drug history; stop/reassess lithium before attributing the picture to an adenoma.

## 8. Mild up-PTH state

### WHAT YOU SEE

Panel labeled MILD with up-PTH, 'mimics primary HPT'.

### WHAT IT ENCODES

Mild CaSR inactivation (FHH) and lithium effect both produce a deceptively benign picture: mildly elevated calcium with normal-to-mildly-high PTH, closely resembling early/mild primary hyperparathyroidism. Distinguishing them rests on urinary calcium and clinical context (family history, age of onset, medications) rather than the serum calcium/PTH pair alone.

### BOARD TRAP

A stem concludes 'high-normal PTH with high calcium = primary hyperparathyroidism' without further testing — WRONG. The same labs occur in FHH and lithium use. The discriminator: order a 24-h urinary calcium / Ca-Cr clearance ratio and review medications before committing to primary HPT.

## 9. Mg Goldilocks panels

### WHAT YOU SEE

Three top-right panels: mild low Mg, severe low Mg, and high Mg.

### WHAT IT ENCODES

Magnesium has a biphasic, 'Goldilocks' effect on PTH. MILD hypomagnesemia stimulates PTH secretion (like low calcium). SEVERE hypomagnesemia paradoxically BLOCKS PTH release and causes end-organ PTH resistance, yielding functional hypoparathyroidism and refractory hypocalcemia. HIGH magnesium activates the CaSR and suppresses PTH. So both too-little and too-much magnesium can lower PTH.

### BOARD TRAP

A stem assumes 'low magnesium always raises PTH' — WRONG. Only MILD hypomagnesemia raises PTH; SEVERE hypomagnesemia suppresses it. The discriminator is severity: the curve is U-shaped, and severe Mg depletion behaves opposite to mild depletion.

## 10. Severe low Mg / PTH suppressed

### WHAT YOU SEE

Panel: severe low Mg with PTH suppressed and end-organ resistance.

### WHAT IT ENCODES

Severe hypomagnesemia (often from PPIs, alcoholism, diarrhea, or aminoglycosides) impairs BOTH PTH secretion and PTH action at bone/kidney, producing hypocalcemia that will not correct with calcium alone. The fix is to replete MAGNESIUM first; once magnesium is restored, PTH secretion and responsiveness recover and calcium normalizes.

### BOARD TRAP

A patient with hypocalcemia that fails to respond to IV calcium is given ever-larger calcium doses — WRONG. Suspect and correct SEVERE hypomagnesemia first. The discriminator: refractory hypocalcemia (especially with low PTH despite low Ca) points to low magnesium; calcium will not respond until Mg is replaced.

## 11. High Mg mimics low Ca

### WHAT YOU SEE

Panel: high Mg figure, noting Mg mimics low Ca and high Mg can suppress PTH.

### WHAT IT ENCODES

Hypermagnesemia activates the CaSR (magnesium is a CaSR agonist), suppressing PTH and lowering calcium. The classic setting is obstetric MgSO<sub>4</sub> for preeclampsia/tocolysis, where rising magnesium causes loss of deep tendon reflexes (first sign), then respiratory depression and cardiac effects. Treatment of symptomatic hypermagnesemia is IV calcium gluconate (direct antagonist) plus stopping the source.

### BOARD TRAP

A stem expects high magnesium to RAISE PTH (treating it like low calcium) — WRONG. High magnesium ACTIVATES the CaSR and SUPPRESSES PTH, lowering calcium. The discriminator: magnesium and calcium act on the same receptor, so hypermagnesemia mimics hypercalcemia at the CaSR (PTH down), even though it lowers serum calcium.

## 12. ADH gain of function

### WHAT YOU SEE

Right-side ADH panel marked 'gain of function', dial thinks Ca is high.

### WHAT IT ENCODES

Autosomal dominant hypocalcemia (ADH, also called type 1 with CaSR gain-of-function) is an ACTIVATING CaSR mutation. The over-sensitive receptor perceives normal calcium as high, so it suppresses PTH and drives renal calcium wasting — yielding hypocalcemia with inappropriately low/normal PTH AND hypercalciuria. It is the mirror image of FHH. Treatment is cautious; calcimimetics worsen it, and recombinant PTH may be used.

### BOARD TRAP

A patient with ADH and symptomatic hypocalcemia is treated with aggressive calcium and vitamin D to normalize serum calcium — WRONG. Because the renal CaSR is overactive, this dramatically worsens HYPERCALCIURIA, causing nephrocalcinosis, stones, and renal impairment. The discriminator: in ADH, treat only to relieve symptoms (target low-normal calcium); PTH is LOW, not high, so this is NOT ordinary hypoparathyroidism management.



### 13. Summary dials (FHH/ADH/Li)

#### WHAT YOU SEE

Bottom comparison dials: FHH normal/low urine Ca, ADH high urine Ca, Li shifted set point.

#### WHAT IT ENCODES

Urinary calcium is the key discriminator among CaSR disorders. FHH (inactivating) = hypercalcemia with LOW urine calcium (Ca/Cr clearance  $<0.01$ ). ADH (activating) = hypocalcemia with HIGH urine calcium. Lithium shifts the setpoint upward, mimicking FHH/primary HPT with variable (often low) urine calcium. Pairing serum calcium with urine calcium and PTH localizes the lesion.

#### BOARD TRAP

A stem pairs an inactivating CaSR phenotype with high urine calcium (or an activating one with low urine calcium) — WRONG; that is backwards. FHH (inactivating) = high serum Ca / LOW urine Ca; ADH (activating) = low serum Ca / HIGH urine Ca. The discriminator: the two are mirror images, and the urine calcium always travels with the direction of CaSR activity (more active receptor  $\rightarrow$  more urinary calcium loss).

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